



Government of Nepal  
Ministry of Urban Development

Central Level Project Implementation Unit

↑ Related with Housing, Settlement, Public Building and Urban Infrastructure

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# PRIVATE HOUSING RECONSTRUCTION

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April 21<sup>st</sup>, 2017

# Activities

- Around 626695 Private Houses are to be constructed in 14 severely affected district out of which 49530 houses have started construction and around 22000 houses are already completed
- Training for around 2570 Engineering Technicians about Earthquake resistant Technology and Inspection Manual Training ( through IDA fund) is being conducted
- Mason Training in Dolakha, Dhading and Nuwakot is conducted through IDA fund and around 300 mason is already trained

# District wise status of Housing Reconstruction

| Sno | District        | No of Benificaries | No of Housing Starting Construction | No of Application for Inspection of 2nd Tranche | No of Houses Certified for 2nd Tranche | No of Houses Non-compliant to Building Codes | No of Application Received for 3rd Tranche | No of Houses Certified for 3rd Tranche | No of Houses Non-compliant to Building Codes | No of Houses Completed |
|-----|-----------------|--------------------|-------------------------------------|-------------------------------------------------|----------------------------------------|----------------------------------------------|--------------------------------------------|----------------------------------------|----------------------------------------------|------------------------|
| 1   | SINDHUPALCHOK   | 78537              | 7232                                | 2167                                            | 1092                                   | 129                                          | 114                                        | 45                                     | 4                                            | 1525                   |
| 2   | DHADING         | 70581              | 4804                                | 3418                                            | 999                                    | 290                                          | 362                                        | 48                                     | 0                                            | 5002                   |
| 3   | KABHREPALANCHOK | 67731              | 5338                                | 4321                                            | 2605                                   | 340                                          | 128                                        | 56                                     | 5                                            | 1758                   |
| 4   | NUWAKOT         | 65773              | 4618                                | 6409                                            | 2691                                   | 870                                          | 79                                         | 79                                     |                                              | 5065                   |
| 5   | GORKHA          | 58503              | 4865                                | 3657                                            | 930                                    | 396                                          | 0                                          | 0                                      | 0                                            | 0                      |
| 6   | DOLAKHA         | 51940              | 5887                                | 4856                                            | 2003                                   | 181                                          | 384                                        | 200                                    | 10                                           | 2890                   |
| 7   | RAMECHHAP       | 43609              | 4184                                | 2970                                            | 2075                                   | 123                                          | 332                                        | 104                                    | 21                                           | 2084                   |
| 8   | KATHMANDU       | 42500              | 818                                 | 299                                             | 85                                     | 39                                           | 0                                          | 0                                      | 0                                            | 160                    |
| 9   | SINDHULI        | 34269              | 2907                                | 1123                                            | 412                                    | 117                                          |                                            |                                        |                                              | 1418                   |
| 10  | MAKAWANPUR      | 30238              | 2939                                | 1739                                            | 524                                    | 1                                            | 33                                         | 33                                     |                                              | 0                      |
| 11  | BHAKTAPUR       | 26066              | 668                                 | 247                                             | 40                                     | 32                                           | 0                                          | 0                                      |                                              | 181                    |
| 12  | LALITPUR        | 25893              | 676                                 | 39                                              | 10                                     | 1                                            | 0                                          | 0                                      |                                              | 319                    |
| 13  | OKHALDHUNGA     | 19819              | 3606                                | 1955                                            | 490                                    | 147                                          | 140                                        | 0                                      |                                              | 288                    |
| 14  | RASUWA          | 11236              | 992                                 | 694                                             | 435                                    | 675                                          | 68                                         | 2                                      | 0                                            | 1466                   |
|     | Total           | <b>626695</b>      | 49534                               | 33894                                           | 14391                                  | 3341                                         | 1640                                       | 567                                    | 40                                           | 22156                  |

# District wise Technical Personnel

| स्वीकृत दरबन्दी भन्दा कम पदस्थापन भएका जिल्लाहरूमा संख्या थप गर्नुपर्ने विवरण |                |                 |              |                 |                    |              |                 |                             |              |                 |        |
|-------------------------------------------------------------------------------|----------------|-----------------|--------------|-----------------|--------------------|--------------|-----------------|-----------------------------|--------------|-----------------|--------|
| क्र.स.                                                                        | DLPIU को नाम   | स्वीकृत दरबन्दी |              |                 | हाल कार्यरत संख्या |              |                 | जिल्ला थप गर्नुपर्ने संख्या |              |                 | कैफियत |
|                                                                               |                | ईन्जिनियर       | सब-ईन्जिनियर | अ. सब-ईन्जिनियर | ईन्जिनियर          | सब-ईन्जिनियर | अ. सब-ईन्जिनियर | ईन्जिनियर                   | सब-ईन्जिनियर | अ. सब-ईन्जिनियर |        |
| १                                                                             | गोरखा          | १२६             | ६५           | ६०              | ११३                | ५०           | ५७              | १३                          | १५           | ३               |        |
| २                                                                             | धादिङ          | १५४             | ५१           | ४६              | ११९                | ३६           | ४१              | ३५                          | १५           | ५               |        |
| ३                                                                             | नुवाकोट        | १५०             | ६६           | ६१              | १०८                | ५२           | ५७              | ४२                          | १४           | ४               |        |
| ४                                                                             | रसुवा          | ३३              | २१           | १८              | २७                 | १३           | १०              | ६                           | ८            | ८               |        |
| ५                                                                             | काठमाण्डौ      | ६१              | ७            | ०               | ६१                 | ७            | ०               | ०                           | ०            | ०               |        |
| ६                                                                             | भक्तपुर        | ३१              | ४            | ०               | ३१                 | ४            | ०               | ०                           | ०            | ०               |        |
| ७                                                                             | ललितपुर        | ३३              | २३           | १८              | ३३                 | २३           | १८              | ०                           | ०            | ०               |        |
| ८                                                                             | काभ्रेपलान्चोक | १३९             | ८०           | ७५              | १२९                | ७४           | ७३              | १०                          | ६            | २               |        |
| ९                                                                             | सिन्धुपाल्चोक  | १६५             | ७३           | ६८              | ११२                | ५७           | ५८              | ५३                          | १६           | १०              |        |
| १०                                                                            | मकवानपुर       | ७६              | ४०           | ३५              | ५१                 | २७           | ३४              | २५                          | १३           | १               |        |
| ११                                                                            | दोलखा          | १११             | ५२           | ४८              | ८०                 | ४२           | ४८              | ३१                          | १०           | ०               |        |
| १२                                                                            | रामेछाप        | ९५              | ५०           | ४५              | ७४                 | ३४           | ४४              | २१                          | १६           | १               |        |
| १३                                                                            | सिन्धुली       | ८९              | ५५           | ५०              | ६४                 | ४७           | ४७              | २५                          | ८            | ३               |        |
| १४                                                                            | ओखलढुंगा       | ६८              | ५४           | ५०              | ५९                 | ३८           | ४८              | ९                           | १६           | २               |        |
| १५                                                                            | SRIU           | ५               | १            | ०               | ५                  | ०            | ०               | ०                           | १            | ०               |        |
| १६                                                                            | CLPIU          | १०              | ६            | ०               | १०                 | ५            | ०               | ०                           | १            | ०               |        |
|                                                                               | जम्मा          | १३४६            | ६४८          | ५७४             | १०७६               | ५०९          | ५३५             | २७०                         | १३९          | ३९              |        |

# Activities

- Training for around 20000 existing mason is being conducted in earthquake affected district through different Pos with approval from MOUD/DUDBC and MOUD-CLPIU have started training for around 5000 Existing Mason in Nuwakot, Dolakha and Dhading through IDA funding
- CLPIU targets to produce around 50000 ( 30000 Mason +20000 Semi Skilled Mason) for Housing construction through Head Mason training and on the job Training with construction of model Houses in Gap VDC in affected district through IDA Source and have already sent request to NRA for necessary budget and programs
- Awareness Program to House-owner and community through IEC material, interactive session, door to door service through DLPIU Engineers and through other telecommunication medium
- Volume II for alternative technologies, more design on RCC Structure

# Activities

- Draft building Bylaws of Traditional settlement has approved by the cabinet of Minister and is a stage of being implemented
- Cabinet decided to enforce National Building code in all Municipality and then VDCs which have more than 1000 Households

# Constraints in housing reconstruction

- Regular and adequate funding for reconstruction activities with high prioritization from Government, Donor Agencies and Partner Organization
- Availability and quality of Construction Mason & Engineering technician
- Supply chain of Construction material
- Quality of construction material
- Various Typology of building constructed
- Availability of Inspection forms in the field
- Duplication of activities
- Non compliance in construction work
- Grievance and its Handling
- Effective supervision and monitoring of housing construction and activities performed by Partner Organization

# Few...Non Compliance issues

## **RCC Houses without minimum Criteria of MRT codes**



# ... Non Compliance issues

- Mixed Structure



# ... Non Compliance issues

- Thickness and Components of Horizontal Bands & vertical bars



# ... Non Compliance issues

L shaped building



# ... Non Compliance issues

2 Story building in Mud Mortar



# Signs of Good Practices

## Build Back Better



# Built Back Better



# ... Built Back Better



# Way Forward

- Formulation of More type -design based on typical housing construction pattern in the field
- Design Catalogue Volume II for Alternative Technology
- Mason Training and OJT Training program in Gap VDCs of all affected districts
- Correction and Exception manual
- Flexible Inspection forms
- House-owner orientation programs
- Regular orientation programs to technicians
- Set up of material storing temporary warehouses in affected areas through NRA and National Trading
- Addressing of Grievances

Thank you !!